

EDUCATION

B.Tech, Computer Science and Engineering

CGPA: 9.4/10.0

Indian Institute of Technology, Kanpur

2017 – 2021

AISSCE (XII), Sunbeam English School, CBSE

96.4%, 2017

WORK EXPERIENCE

Software Development Engineer II

Hyderabad

Microsoft – Azure Container Apps

Jul’24 - Present

- Designed **AI-powered SRE agents** for ACA incident diagnostics, reducing P50 time-to-mitigate by **97%** (185→6 min); evolving toward **autonomous auto-repair**
- Built **Session Management APIs** and fixed async execution bugs for ACA Sessions, improving stability for **~300 customers**; enabled billing optimization for Fabric team
- Expanded **Azure Functions on ACA** — keys management, KEDA rules, drain, tracing for **~1,322 apps**; enabled **Foundry Control Plane** for ACA agents
- Led **Dapr AKS Extension** rollouts (v1.15, v1.16) across all Azure regions; remediated CVEs for **~400 clusters**; integrated CodeQL for pipeline security
- Integrated **Azure Key Vault** with Managed Environment Storage API, securing **~27K daily API calls** by eliminating raw connection strings
- On-call DRI: resolved **200+ incidents** (27+ CRIs), outperforming team baselines; mentored new hires; presented at **IDC Pinnacle**

Software Development Engineer

Hyderabad

Microsoft – Azure Container Apps / Dapr

Jul’21 - Jul’24

- Contributed to **Dapr** (CNCF OSS) — upgraded MQTT, CosmosDB, Config API, Redis, PostgreSQL to stable; implemented Workflow **cascade terminate/purge** across SDKs
- Architected **internal fork** across 5 repos with CI/CD and release pipelines; automated releases within **<1 day SLA**
- Migrated to **OneBranch/1ES** with Ev2 deployment; added multi-arch, Mariner and distroless images; 1P registry migration
- Enabled **airgap install**, winget and MSI for Dapr CLI; added bulk subscribe in Azure Event Hubs; resolved customer issues on GitHub and Discord
- Added **Dapr Workflow** performance tests identifying P0 bugs; published combined coverage reports for conformance and certification tests
- Delivered nightly builds for ACA; contributed to AKS Extension GA requirements; resolved issues for Container Apps team

Quantitative Research Intern

May’20 - Jul’20

JP Morgan & Chase – E-Trading, Mumbai

- Built price prediction models using **LSTM**, neural networks, and regression with univariate feature selection
- Improved predictive power by **40%** for HK names via feature standardization
- Implemented L1 and L2 features for optimizing model parameterization

RESEARCH EXPERIENCE

Data Race Detection, Task-Parallel Programs

Jun’19 - Aug’21

Supervisor: Prof. Swarnendu Biswas, IIT Kanpur

- Designed novel algorithm **Tasker** integrating vector clocks with tree-based techniques; used **LLVM** pass for memory instrumentation
- Achieved **1.48X speedup** over PTRacer baseline on 128GB Intel Xeon system
- “Efficient Data Race Detection of Structured Task Parallel Programs Using Vector Clocks” — S. Kumar, A. Agrawal, S. Biswas | Submitted, ISCGO 2022

TECHNICAL SKILLS

Languages: Go, Python, C#/.NET, JavaScript/TypeScript, C/C++

Cloud: Azure (ACA, AKS, Functions, Key Vault, Event Hubs, CosmosDB)

Infra: Kubernetes, Docker, Dapr (CNCF), Helm, Container Registries

Tools: OneBranch, Ev2, ADO Pipelines, GitHub Actions, Geneva, CodeQL

Domains: Distributed Systems, AI/LLM Agents, CI/CD, Microservices, SRE

ACADEMIC PROJECTS

- Java Compiler** (CS335) — Designed lexer, parser, and 3-address code generation using PLY; support for classes, functions, and interfaces
- GemOS** (CS330) — Implemented mmap, pagefaults, lazy allocation, and copy-on-write fork syscalls
- Machine Learning** (CS771) — CNN image classification using Keras, SGD, multi-label Bonsai recommender system
- Cipher Decoder** (CS641) — Differential cryptanalysis of 3-DES; decryption of Caesar, Vigenere ciphers
- SAT Solver** (CS202) — DPLL algorithm; encoded diagonal sudoku in DIMACS propositional logic
- MERN Mobile App** (CS252) — Full-stack app with React Native, Node.js, MongoDB, and secure login

ACHIEVEMENTS & OTHERS

- RockStar Award**, Microsoft (’22, ’23) for exceptional contributions
- Academic Excellence Award** (×2), IIT Kanpur (’17-18, ’18-19)
- Ram Prakash Chopra Memorial Scholarship
- JEE Advanced **AIR 715** | KVPY Scholar | NSE Physics **Top 1%**
- Academic Mentor, Counselling Service, IIT Kanpur (’18-19)
- Secretary, Dramatics Club, IIT Kanpur (’18-19)

RELEVANT COURSES

Operating Systems

Compiler Design

Computer Networks

Adv. Algorithms

Machine Learning

Modern Cryptology

Parallel Computing

NLP